

Turnstile

Real-time event check-in system

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Project Description

Turnstile solves one problem: making sure the same ticket can't get two people through the door, while giving organizers a live, accurate picture of who's actually in the building. Organizers create an event and set a capacity; attendees register and receive a unique, one-time-use QR code; organizers scan attendees in at the door via a camera-based scanner; and a live dashboard shows check-ins as they happen, in real time, across multiple scanning stations at once.

The two hard guarantees — no over-capacity registration, and no duplicate check-ins, even under concurrent load from multiple scanner stations — are enforced at the database layer itself (SQLite, WAL mode), not by application-level locks, and are verified by a dedicated concurrency-proof script that runs the backend as two separate OS processes hitting the same database file.

Node.js + Express + TypeScript

SQLite (node:sqlite)

Socket.io

JWT auth

QR codes (qrcode / html5-qrcode)

Gemini 2.5 Flash-Lite

React + TypeScript + Vite

Tailwind CSS v4

Features Built

- **Role-based auth** — JWT-based signup/login with separate organizer and attendee roles, gating what each can see and do.
- **Event creation & capacity-bounded registration** — organizers set a capacity; registration is rejected once it's full, enforced atomically at the DB level under concurrent requests.
- **Per-attendee QR tickets** — each registration gets its own signed, one-time-use QR code, generated server-side.
- **Camera-based check-in scanner** — live camera QR scanning with instant valid/duplicate/invalid feedback, plus a manual-entry fallback for testing or camera-less stations.
- **Offline-first scanning** — a flaky venue network doesn't drop check-ins: scans queue locally when offline and sync automatically in the background once connectivity returns.
- **Duplicate check-in prevention** — guaranteed at the database layer, not in application code, and covered by a multi-process concurrency proof.
- **Live organizer dashboard** — real-time registered/checked-in/spots-left stats and a live scan feed via Socket.io, updating instantly as check-ins happen at the door.
- **AI-powered insights** — ask natural-language questions about an event's check-in data (Gemini 2.5 Flash-Lite), with a graceful fallback to raw computed stats if no API key is configured.
- **CSV export** — full attendee list with check-in timestamps, one click from the dashboard.

Screenshots

Turnstile

Event check-in system.

SIGN INREGISTER

ACCOUNT TYPE

ORGANIZER

ATTENDEE

NAME

EMAIL ADDRESS

operator@turnstile.app

PASSWORD

Initialize Identity

Sign in / register — single tabbed auth screen with an organizer/attendee account-type picker.

TURNSTILE

Search events...

SYSTEM: ONLINE

SIGNED IN AS

Org gnv3c

ORGANIZER

Browse Events

Create Event

Scanner

LOGOUT

Explore Events

Discover and register for upcoming events. Spots fill up quickly, so reserve your place in advance.

CREATE EVENT

REGISTRATION OPEN

1 SEPT 2026, 23:30

E2E Event

Capacity1/3 filled

Browse Events (organizer view) — capacity fill bar and registration-status badge per event.

TURNSTILE

Search...

SYSTEM: ONLINE

SIGNED IN AS
Org gnv3c
ORGANIZER

Create New Event

Set up the event's parameters. Attendees will be able to register once it's live.

Browse Events

Create Event

Scanner

EVENT NAME

Demo Launch gnv3c

DATE & TIME

22/08/2026, 17:59

CAPACITY

50

CREATE EVENT →

LOGOUT

Create Event — organizer sets name, date/time, and capacity.

TURNSTILE

Search...

SYSTEM: ONLINE

SIGNED IN AS
Org gnv3c
ORGANIZER

Demo Launch gnv3c

LIVE TELEMETRY ACTIVE

22/08/2026, 17:59:00

EXPORT CSV

Browse Events

Create Event

Scanner

REGISTERED

1/50

CHECKED IN

0

0.0%

SPOTS LEFT

49

AI ANALYSIS

How many spots are left?

ASK

How many people have checked in so far?
What percentage of registered attendees are no-shows?
What time did check-ins peak?
How many spots are left?

There are 49 spots left for Demo Launch gnv3c.

Generated: 23:29:13

Live Scan Stream

TIMESTAMP	ATTENDEE	STATUS
Waiting for check-ins...		

Attendees

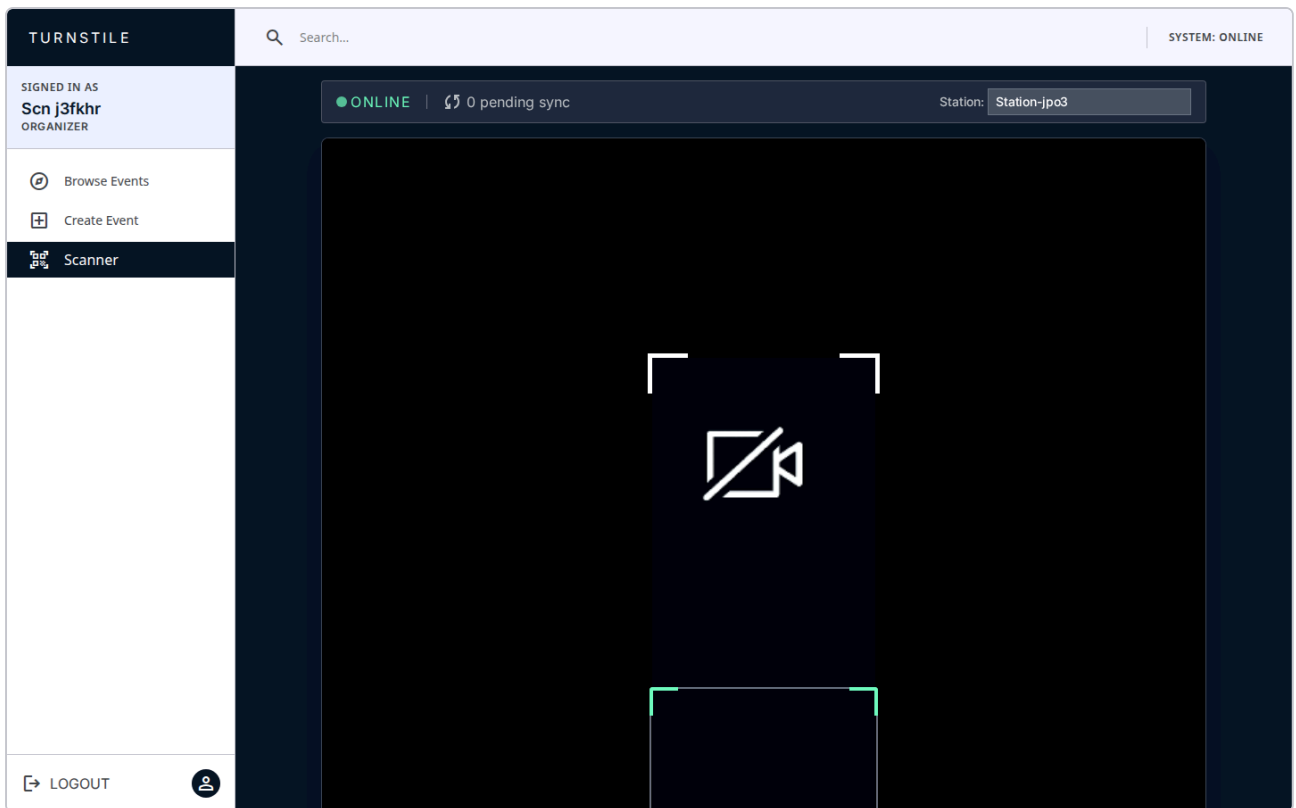
Att gnv3c (att-gnv3c@example.com)

not checked in

LOGOUT

Live organizer dashboard — real-time stats, live scan stream over Socket.io, and the AI insights panel answering a natural-language question about the event.





Check-in scanner — live camera viewfinder with online/offline status and pending-sync counter for offline-first scanning.